



Heat wave and the risk of intimate partner violence

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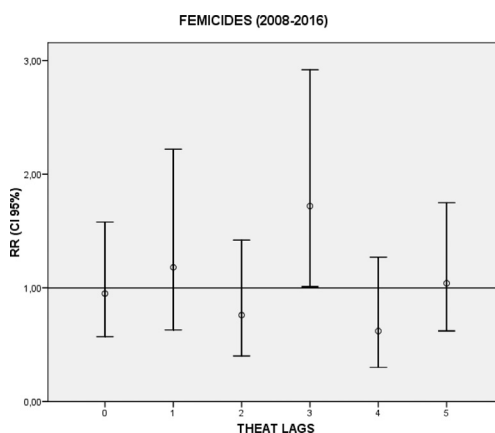
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HIGHLIGHTS

- The risk of intimate partner femicides increases three days after the heat wave.
- The risk of police reports increases one day after the heat wave.
- The risk of help line calls increases five days after the heat wave.
- Heat waves are associated with an increase in intimate partner violence.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 18 May 2018

Received in revised form 28 June 2018

Accepted 29 June 2018

Available online 6 July 2018

Editor: Scott Sheridan

Keywords:

Heat wave

Intimate partner violence

Femicides

ABSTRACT

Background: A high number of women report experiencing intimate partner violence (IPV). It is of utmost importance to identify possible factors that precipitate IPV and incorporate them into police protocols for evaluating IPV risk. Scientific evidence shows that environmental temperature is associated with a risk of violent behavior.

Objectives: To analyze the effect and impact of heat waves on the risk of IPV.

Methods: Ecological, longitudinal time series study. The dependent variables are: intimate partner femicides (IPF), reports of IPV and 016 IPV telephone help line calls in the Community of Madrid from 05/01 to 09/30 in the years 2008–2016. The principal independent variable is the daily maximum temperature in Celsius (Tmax) above the heat wave threshold of 34 °C. A binomial negative regression was used for calls and reports and a Poisson regression was used for IPF. The attributable risk among those exposed (AR%) and the number of attributable cases was calculated for each variable.

Results: The risk of IPF increased three days after the heat wave, [RR(IC95%):1.40(1.00–1.97)], police reports of IPV increased one day after [RR(IC95%):1.02(1.00–1.03)] and help line calls increased five days after [RR(IC95%):1.01(1.00–1.03)]. The AR% was 28.8% (IC95%: 0.3%–49.2%) for IPF, 1.7% (IC95%:0.3%–3.1%) for police reports and 1.43% (IC95%:0.1%–2.8%) for help line calls.

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Conclusions: Our results suggest that heat waves are associated with an increase in IPV. The effect of an increase in IPV is delayed in time, with differences according to the violence indicators analyzed.

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1. Introduction

According to the last World Report on Homicide of the United Nations, 43,600 women were killed by their intimate partners or family members in 2012 (UNODC, 2013). In the European Union (EU) it is estimated that 22% of the women that have or have had a partner have experienced physical and/or sexual violence, and 43% have experienced psychological violence by a current or past intimate partner (FRA, 2014). In Spain, according the last Macrosurvey on Gender Violence 2015, the 13% of the women had suffered everlife physical and/or sexual intimate partner violence (IPV) (Delegación de Gobierno para la Violencia de Género, 2015, Macroencuesta).

Despite the fact that only 30% of the European women (FRA, 2014) and the 26.8 of the Spanish women (Delegación de Gobierno para la Violencia de Género, 2015, Macroencuesta) exposed to this type of violence file a police report against their aggressor (FRA, 2014), in Spain in 2017 there were 125,769 reports of intimate partner violence (IPV) (including 20,044 in the Community of Madrid). Also in 2017, 49 women were killed by their partners, only 11 (22.4%) of whom had filed a police report against their aggressor (Ministerio de Sanidad Servicios Sociales e igualdad, 2015; <http://estadisticasviolenciagenero.msssi.gob.es/>). These figures, which are repeated year after year in different European countries (FRA, 2014; García-Moreno et al., 2006) highlight the need for social innovation to address this grave social problem.

While the roots of IPV are clearly linked to gender inequalities, in recent decades, there has been intense research to identify factors that increase a woman's risk of experiencing IPV (Abramsky et al., 2011; López-Ossorio et al., 2017; Sanz-Barbero et al., 2018) and, in the most extreme terms, of being killed by her partner (Beyer et al., 2013; Sanz-Barbero et al., 2016; Stöckl et al., 2013). These factors, as pointed out by Heise's Integral Ecological Model (Heise, 1998), interact on different levels ranging from individual characteristics - age, birthplace, socioeconomic position, etc. - to other social and structural circumstances such as gender inequality and women's political participation (Bond et al., 2010; Gressard et al., 2015; Palma-Solis et al., 2008; Sanz-Barbero et al., 2015).

Given the high number of women who report IPV and request support, it is of vital importance to open new lines of inquiry that permit continued advances in the identification of possible precipitating factors of IPV and their incorporation in police protocols to evaluate the risk of recurrence (López-Ossorio et al., 2016). One of these areas is related to exploring the relationship between IPV and environmental variables more deeply, given its growing presence in research on interpersonal and collective violence (Levy et al., 2017; Michel et al., 2016; Schinasi and Hamra, 2017).

Although it would seem that a behavior as socially rooted as IPV could never be brought about by variables of an environmental nature, scientific evidence shows that exposure to environmental temperature is associated with the risk of violent behavior (Schinasi and Hamra, 2017). In particular, sexual aggression (McLean, 2007), assaults and robberies (Lemon et al., 2017; Michel et al., 2016; Schinasi and Hamra, 2017) have been associated with temperature, although there is controversy over whether this association is linear or curvilinear (Bushman et al., 2005). In the case of IPV, an association has been identified between temperature and an increase in police reports of IPV (Rotton and Cohn, 2001; Auliciems and DiBartolo, 1995; Cohn, 1993), in emergency telephone calls (Rotton and Cohn, 2001; Lebeau, 1994), and in requests for access to domestic violence shelters (Michael and Zumpe, 1986).

The objective of this study is to analyze the effect of daily maximum temperatures during heat waves on the risk of intimate partner femicides (IPF), police reports and calls to the 016 telephone help line in the Community of Madrid, and to estimate the number of cases attributable to this type of violent behavior produced during heat waves.

2. Methods

2.1. Study design

The study is an ecological longitudinal time series study.

2.2. Description of the variables

The dependent variables analyzed include IPFs, IPV police reports and calls to the 016 telephone help line in the Community of Madrid, during the period from May 1st to September 30th in the years 2008–2016. The data sources of the dependent variables include:

- Information about the number of daily calls to the IPV telephone help line in the Community of Madrid during the study period was provided by the Government Delegation for Gender Violence (Ministry of Health, Social Services and Equality).
- The daily number of IPV reports filed before the Gender Violence Courts and the number of IPFs that occurred in the Community of Madrid were provided by the Integral Monitoring System for Cases of Gender Violence (VioGen) of the Ministry of the Interior during the study period mentioned above.

The variable used to account for the impact of heat on the dependent variables was daily maximum temperature (T_{max}). This information was provided by the State Meteorological Agency. According the Ministry of Health, Social Services and Equality's protocol (Díaz et al., 2015; Ministerio de Sanidad, Servicios Sociales e Igualdad, 2017), a heat wave occurs when the daily T_{max} surpasses the threshold of 34 °C in the Observatory in Madrid. Therefore, the principal independent variable is constituted by T_{heat} values defined as:

$$T_{heat} = 0 \text{ si } T_{max} < 34^{\circ} \text{C}$$

$$T_{heat} = T_{max} - 34 \text{ si } T_{max} > 34^{\circ} \text{C}$$

Based on the values of T_{heat} , and with the objective of analyzing whether the effect of high temperatures on the dependent variables could have a lag of one to five days, the variables were calculated with time lags T_{heat1} , T_{heat2} , T_{heat3} , T_{heat4} y T_{heat5} , respectively. The choice of lags of up to 5 days is based on how high temperatures (heatwaves) affect health indicators such as morbi-mortality at a biological level (Díaz et al., 2002) and the impact in mental illnesses such as neurodegenerative (Linares et al., 2016). In both cases the delays chosen are up to 5. On the other hand, the papers of Katerndahl et al. (Katerndahl et al., 2010), indicate that the delay between stress reactions and alcohol abuse with femicides are usually delayed up to 3 days.

2.3. Statistical analyses

Given that in the case of the variables "IPV police reports" and "calls to the 016 IPV telephone help line" there is overdispersion - greater

variance than would be expected on the assumption of independence and homogeneity which underlies Poisson distribution - , a binomial negative regression was used to calculate the coefficients that permit determining the value of the RR. In the case of IPV femicides, the generalized linear model methodology with the Poisson regression link is used.

The following covariables were included in the analysis, in order to control for the trend and seasonalities of the series, as well as the lags in the T_{heat} mentioned before:

- Sine and Cosine functions of 90 and 120 days to account for three and four month periodicities.
- The trend of the series, using a counter (n_1), which is 1 for the first day of the series, 2 for the second day, and so on, successively.
- Days of the week, using dummy variables.
- Summer vacation period: July 1st through August 31st.
- Autoregressive character of the series by introducing an autoregressive term order 1 (AR1).
- The modeling process was carried out in reverse order manually, eliminating the variables, by procedure step-step until $P < 0.05$

The increase for the calculation of RR for T_{heat} was 1 °C above the threshold ($T_{\text{max}} 34$ °C). The proportional attributable risk among the exposed (AR%) was calculated based on the RR and following Coste and Spira's equation (Coste and Spira, 1991) with CI of 95%, and $\text{AR\%} = (\text{RR} - 1) \times 100 / \text{RR}$. Additionally, the number of IPFs, calls to IPV telephone help line 016 and IPV police reports attributable to heat was calculated by applying the AR to the series of dependent variables, following the methodology commonly used to calculate the attributable impact to extreme temperatures (Carmona et al., 2016).

The statistical software packages used to carry out this analysis were: IBM SPSS Statistics 24 and STATA v11.2.

3. Results

During the months of May through September of 2008–2016, 304 days were declared heat waves with $T_{\text{max}} > 34$ in the Community of Madrid. The average maximum temperature during the heat waves was 35.8 °C (T_{heat} average 1.8 °C; DES: 1.3).

There were 23 cases of IPF in the geographic region in question during the study period, and there were 61,117 calls to the emergency 016 IPV telephone help line. There were 38,088 IPV police reports filed. The 304 days with $T_{\text{max}} > 34$ °C included 5 IPV femicides, 13,998 help line calls and 8906 IPV police reports.

The descriptive statistics of the dependent variables are shown in Table 1.

Once we carried out the regression models (Table 2), we observed that independently of the covariables included in the model, the association between T_{heat} and IPFs occurs with a gap of three days. Three days after a heat wave in Madrid with $T_{\text{max}} > 34$ °C there is a 40% increase in the risk of femicide [$\text{RR}-T_{\text{heat}3}$ (IC95%): 1.40 (1.00–1.97)], as showed in Fig. 1, which corresponds to a proportional attributable risk of 28.8% (IC95%: 0.3%–49.2%). That is, for each one degree in daily maximum

temperature above the threshold of 34 °C, there is an increase in IPF of 28.8% with respect to the average.

In the case of IPV police reports, the association is produced one day after the daily maximum temperature surpasses 34 °C (Fig. 2). In this case, the result of the negative binomial modeling shows that $\text{RR}_{T_{\text{heat}1}}$ is 1.02 (IC95%: 1.00–1.03), which corresponds to an AR% of 1.7% (IC95%: 0.3%–3.1%). For each degree that T_{max} surpasses the 34 °C threshold, IPV police reports increase by 1.7% with respect to the average.

With respect to calls to IPV telephone help line 016, the results show that, independently of the covariables included in the model, calls to the 016 telephone help line increase 5 days after the start of the heat wave (Fig. 3), with an RR associated with a delay of 5 $\text{RR}-T_{\text{heat}5} = 1.01$ (IC95%: 1.00–1.03) which corresponds to an AR of 1.43% (IC95: 0.1%; 2.8%).

3.1. Femicides, police reports and telephone help line calls attributable to the increase in IPV during heat waves

During the 304 days of heat wave that occurred in the study period, the average degrees above the temperature threshold was 1.8 °C, and the total degrees above the threshold temperature of 34 °C was 549.2 °C. If the proportional attributable risk among those exposed is considered -AR%- calculated for each of the dependent variables, for each degree that daily T_{max} surpasses the threshold of 34 °C, the number of cases attributable to the increase in violence during heat waves was 9 (0–16) for IPF; 821(147–1490) for IPV police reports, and 1100 (61–2136) for emergency IPV telephone help line calls. Therefore, 39.1% of IPFs that occurred within the study period and in the geographic context would be attributable to the increase in violence during heat waves, as would 2.2% of the IPV police reports and 1.8% of the 016 telephone help line calls.

4. Discussion

To our knowledge, this is the first study that analyzes the effect of abnormally high temperatures on three different IPV indicators. The results of this work show that, independently of the trend and seasonalities, there is a positive and statistically significant relationship between the different IPV indicators analyzed and T_{max} reached during heat waves. This association is statistically significant, not on the day of temperature increase, but a number of days after T_{max} surpasses 34 °C. This lag is different for each of the dependent variables analyzed. Specifically, the risk of IPF increased three days after a day with abnormally high T_{max} , whereas for IPV police reports this lag was one day and for calls to telephone help line 016 it was five days.

The association between temperature and IPV was described three decades ago. Specifically, an association has been identified between temperature and an increase in demand at domestic violence shelters (Michael and Zumppe, 1986), police reports (Rotton and Cohn, 2001; Auliciems and DiBartolo, 1995; Cohn, 1993) and calls to telephone emergency help lines (Rotton and Cohn, 2001; Lebeau, 1994). The association between IPV and temperature has been described as linear by authors (Anderson et al., 2000; Michael and Zumppe, 1986), and it is supported by the framework of the General Affective Aggression Model (GAAM) (Anderson et al., 1995). According to this model, exposure to

Table 1

Descriptive statistics of the dependent variables during the period May–September 2008–2016, Autonomous Community of Madrid.

Variables	Total			Theat > 0 °C			Theat = 0 °C		
	Sum	Avg	STD	Sum	Avg	STD	Sum	Avg	STD
IPV femicides	23	0.02	0.13	5	0.02	0.13	18	0.02	0.13
Telephone calls IPV helpline —016	61,117	43.75	13.26	13,998	46.05	14.81	47,119	43.91	13.12
IPV police reports	38,088	25.75	8.65	8906	29.3	8.34	29,182	27.02	8.49
Number of days	1377			304			10,732		

Table 2

Effect of the maximum temperature > 34 °C on intimate partner violence, May–September 2008–2016, Autonomous Community of Madrid.

Variables	Telephone calls IPV line - 016-				IPV police reports				IPV femicide			
	RR ^a	P > z	[95%IC]		RR ^a	p	[95%IC]		RR ^b	P > z	[95%IC]	
n1					1.00	0.007	1.00	1.00				
Autoregressive term	1.01	<0.001	1.01	1.01	1.00	<0.001	1.00	1.01				
Theat1					1.02	0.017	1.00	1.03				
Theat3									1.40	0.048	1.00	1.97
Theat5	1.02	0.022	1.00	1.03								
Monday	1.33	<0.001	1.28	1.38	1.12	<0.001	1.06	1.17	3.11	0.028	1.13	8.57
Tuesday	0.95	0.013	0.91	0.99								
Friday	0.96	0.062	0.92	1.00	1.06	0.003	1.02	1.11				
Saturday	0.84	<0.001	0.81	0.88	1.32	<0.001	1.27	1.37	3.48	0.011	1.32	9.14
Sunday	0.93	0.001	0.90	0.97	1.56	<0.001	1.50	1.62				
Sine 120					0.97	0.010	0.95	0.99				
Cosine 90	0.98	0.047	0.96	0.99								
Summer holidays									0.45	0.119	0.16	1.23

^a RR obtained by multivariate binomial negative regression.^b RR obtained by multivariate Poisson regression.

high temperatures produces irritability and stress that lowers the threshold for violence. Other authors (Cohn and Rotton, 1997) claim that there is an inverted curvilinear association and identify greater violence with moderate temperatures than when temperatures are high.

In contrast to prior studies, our study is the first to exclusively analyze the effect of extremely high temperatures and the first to include the effect during the five days after exposure in the model. The results show that the effect of the daily maximum temperature on gender violence variables is especially important on days with high with high temperatures and the following days. This indicates that a functional relationship between temperature and the dependent variables would be compatible with a “J” or linear relationship, not an inverted “U” form as indicated by other authors (Cohn and Rotton, 1997). This relationship between gender violence variables and high temperatures could be explained by the GAAM. Higher temperatures increase aggressiveness, but the effect is not immediate, rather it is delayed in time.

This design allowed us to identify the way in which the risk of IPV increases three days after a day with an abnormally high Tmax. This lag between exposure to an abnormally high Tmax and femicide could be explained by the general effect of heat on violence and by the couple relationship in those cases in which a femicide is preceded by reports of IPV (López-Ossorio et al., in press). Studies by Katerndahl (D. Katerndahl et al., 2010) monitor couples' relationships daily in a group of women exposed to IPV. They analyze the couple dynamic by level of anger, stress, arguments, alcohol intake, etc. Days with physical abuse follow a homogeneous pattern characterized by high stress, arguments and alcohol consumption. This dynamic of crisis extends across three days, with increasingly grave situations of violence. Although we cannot know in our research whether femicides were preceded by IPV or not, it is possible that exposure to abnormally high temperatures could be a stressor that initiates this type of couple dynamic (anger, stress, arguments...) which culminates in IPV three days later, as

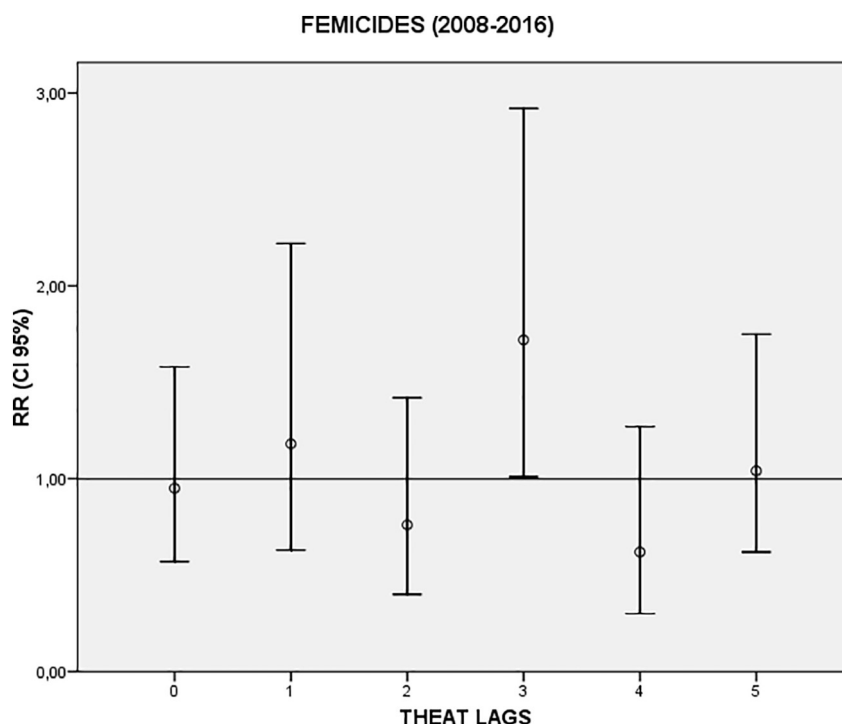


Fig. 1. Relative Risks (RR) corresponding to femicides (2008–2016) according to the Theat lags.

INTIMATE PARTNER VIOLENCE REPORTS (2008–2016)

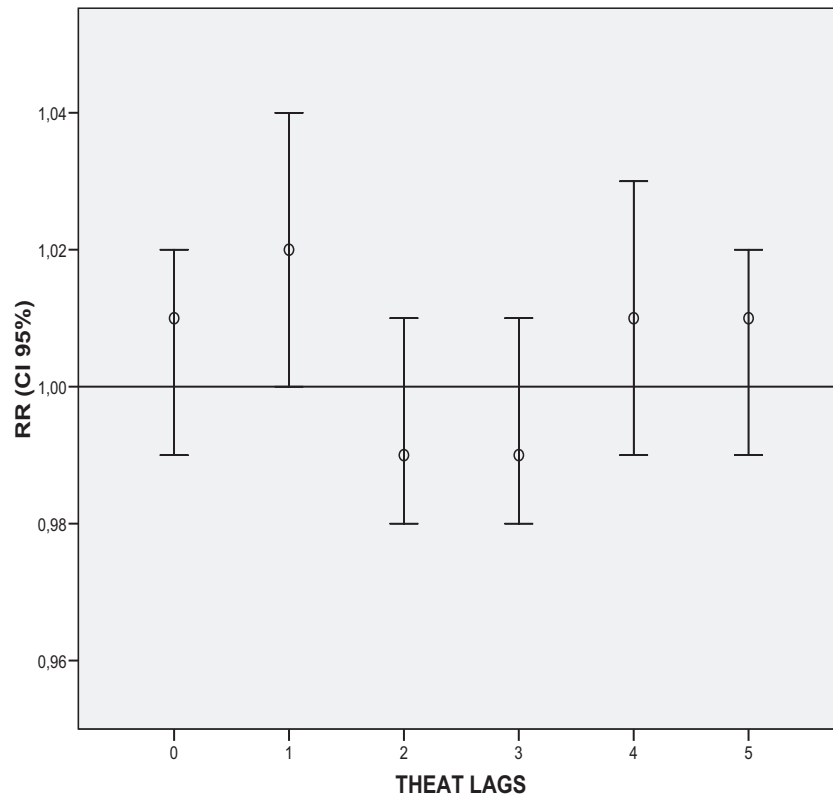


Fig. 2. Relative Risks (RR) corresponding to intimate partner violence reports (2008–2016) according to the Theat lags.

shown in our results. This lag between exposure to abnormally high temperatures and femicide could also indicate that a large part of femicides preceded by IPV are planned acts (Juodis et al., 2014). This provides a margin of time in which to plan public police and individual avoidance strategies once an increased risk is identified by the women and also by the police and social service protocols. In this sense, our study shows that IPV police reports increase one day after exposure to abnormally high temperatures, which could be related to a woman's perception of the gravity of danger with respect to her situation of violence. This is certainly a key finding for the development of measures to guarantee women's safety.

The increase in calls to the 016 telephone IPV help line, five days after the beginning of a heat wave, could indicate that women exposed to IPV use this service once the crisis of violence remits. In this sense, both this work and that of Sanz-Barbero et al. (2018) show that in the context of this study (the Community of Madrid), calls to the 016 telephone line increase on Mondays, possibly when the crisis has passed, while the risk of reports of IPV increases primarily on weekends.

This study should be interpreted considering its strengths and weaknesses. It is important to highlight that, despite the fact that the number of IPV that occurred in the geographic area of the study is less than the number of help line calls and police reports, our results show a statistically significant association between T_{max} and femicides. It is important to note that the fact that all of the IPV indicators analyzed are positively associated with temperature (greater risk of IPV with higher extreme temperatures) gives credibility to our data and indicates the possibility that applying this study design at the national level, with a larger sample size, would provide more precise measures of impact.

Furthermore, the unavailability of individual data does not allow us to analyze the role of alcohol in our study. In the literature, there are studies as conducted by Card and Dahl, 2011 that explains the consumption of some types of alcohol is heavily determined by

temperatures and this consumption would be related with an increase of the IPV.

The analysis of 016 IPV telephone help line calls records the absolute number of daily calls, not the number of callers, therefore it is impossible to identify those women who call more than once per day. Also, IPV police reports is an indicator that represents a fraction of women exposed to IPV. In Spain in 2014, one in three women exposed to IPV reported her aggressor to the police (Delegación de Gobierno para la Violencia de Género, 2015, Macroencuesta), while one third (30%) of women who were killed by their partners in the same year had filed a police report (<http://estadisticasviolenciagenero.msssi.gob.es/>).

Despite the limitations mentioned, the results of this study have important implications for the design of IPV prevention strategies. Firstly, the heat waves can be predicted as part as the monitoring of meteorological systems. The fact that the increase in IPV is delayed in time could permit putting preventive measures into place.

On the other hand, quantifying the impact on the percent of femicides, police reports and help line calls provides a measure of the incidence attributable to heat of these variables. Including temperature in the police risk evaluation protocols and increasing formal resources of support for women exposed to IPV, such as surveillance of women with police protection, police presence in the streets on days predicted with abnormal increase in maximum temperatures and in the days following, could dissuade aggression and minimize the risk of aggression.

It is urgent to promote filing a formal police report among women exposed to IPV, since this opens the door to the Integral Monitoring System for Cases of Gender Violence. Putting IPV telephone help line services on alert and encouraging women to call them on days with abnormally high temperatures is also important in strategies to confront a possible increase in IPV. It would be of special interest to include in the Heat-health actions plans to the women at risk of IPV, as a group of special vulnerability.

CALLS TO TELEPHONE HELP LINE 016 (2008–2016)

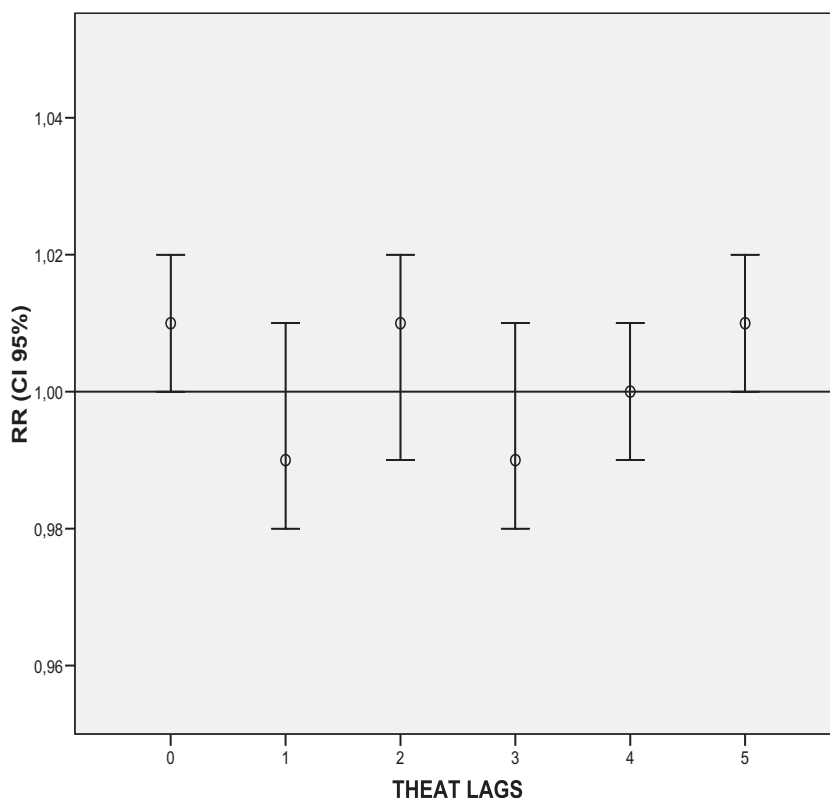


Fig. 3. Relative Risks (RR) corresponding to calls to telephone help line 016 (2008–2016) according to the Theat lags.

5. Conclusions

Our work suggests that extremely high temperatures are associated with an increase in gender violence against women. The effect of this violence is extended in time, with different lag periods according to the violence indicators analyzed. This lag between the increase in temperature and violence could be a key to implementing measures that prevent the fatal consequences of IPV.

Conflict of interest

All authors declare they have no actual or potential competing financial interest.

Disclaimer

The authors declare they have no actual or potential competing financial interests. This article presents independent research. The views expressed are those of the authors and not necessarily those of the Carlos III Institute of Health.

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